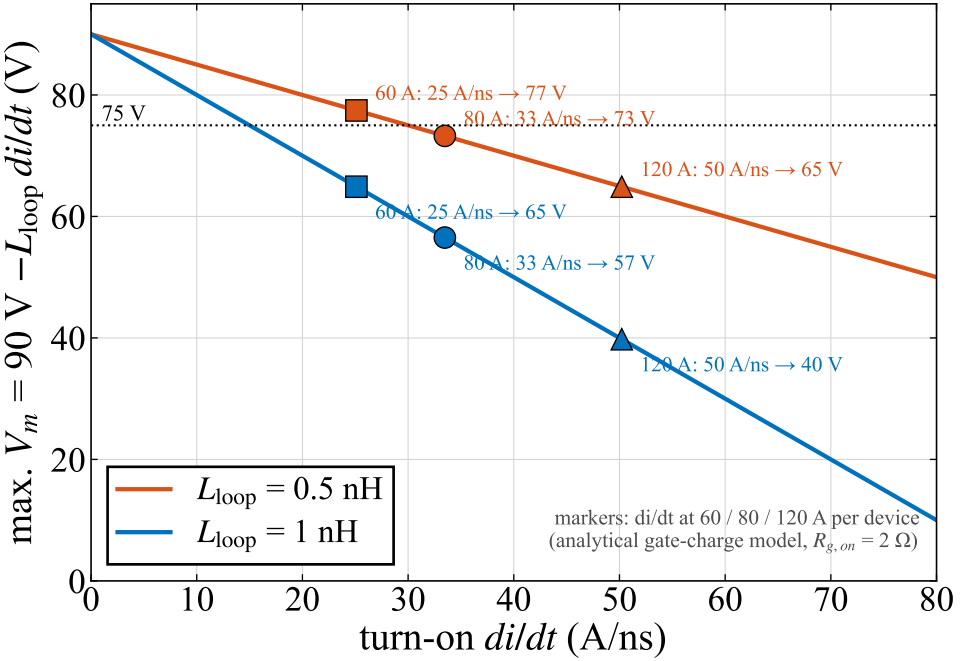
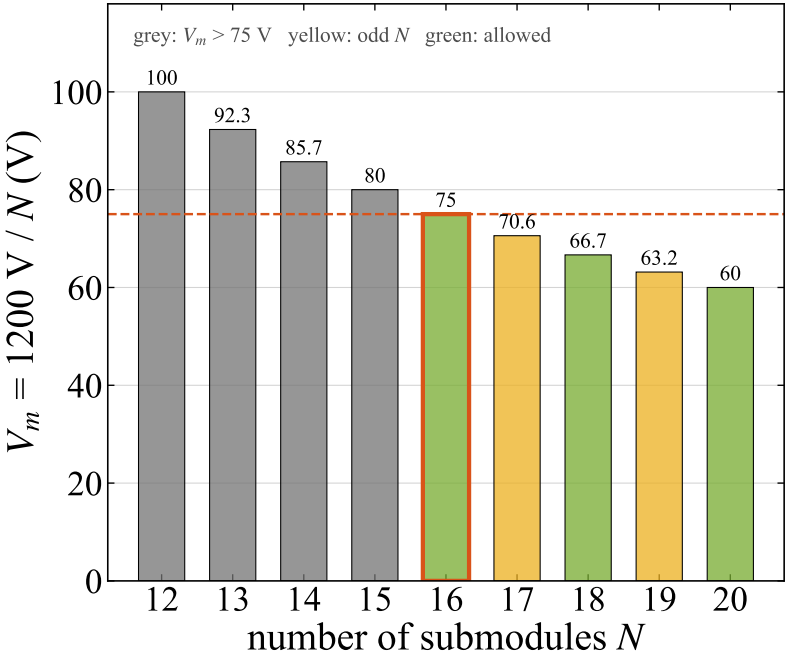

max. $V_m = 90\ \text{V} - L_{\text{loop}}\, di/dt$ (V)
turn-on di/dt (A/ns)
75 V
60 A: 25 A/ns → 77 V
80 A: 33 A/ns → 73 V
120 A: 50 A/ns → 65 V
60 A: 25 A/ns → 65 V
80 A: 33 A/ns → 57 V
120 A: 50 A/ns → 40 V
L_{loop} = 0.5 nH
L_{loop} = 1 nH
markers: di/dt at 60 / 80 / 120 A per device
(analytical gate-charge model, $R_{g,on}$ = 2 Ω)

(a)


grey: V_m > 75 V yellow: odd N green: allowed
$V_m = 1200\ \text{V} / N$ (V)
number of submodules N
100
92.3
85.7
80
75
70.6
66.7
63.2
60

(b)